Venkata Lakshmi Parimala Pasupuleti

Software Engineer · Full Stack · AI-Powered Compliance Features · Python · React · FastAPI · End-to-End Ownership · Financial Services AI · Production

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AgentProbe (live eval toolkit) | Fairfax, Virginia · Open to Onsite NYC

PROFESSIONAL SUMMARY

Full stack software engineer with 7+ years of experience building production web applications, now specializing in **AI-powered features that automate historically manual compliance workflows**. Currently at Intuit owning **financial services AI features end-to-end** - from technical design through implementation and production - with Python/FastAPI backends, React/TypeScript front ends, **relational databases**, and **data modeling** for complex financial data. Built **Leasel**: an AI-powered compliance investigation platform - 8-agent LangGraph workflow with hybrid-search RAG, confidence-gated HITL escalation, and **responsible AI enforced in code** (LLM reasons; human or deterministic logic decides every regulated call) - 93.6% precision across 1,240 test cases. Built **AgentProbe**: open-source evaluation infrastructure for AI features in production. 7+ years at Entrata (100K+ MAU) shipping **full stack features end-to-end** across compliance, income verification, and affordable housing workflows - **frontend architecture, application state, APIs, and data-dense applications** at scale. Product/UX judgment for complex compliance workflows across every role. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026). — github.com/venkatalakshmiparimala/AgentProbe

CORE COMPETENCIES

Full Stack Development | AI-Powered Compliance Features | End-to-End Ownership | Frontend Architecture & Application State | Backend APIs & Data Modeling | Relational Databases | Regulatory Compliance Automation | AI Agents & Agentic Workflows | RAG Pipelines & LLM Features | Evals & Guardrails | Python & FastAPI | React & TypeScript

EXPERIENCE

Intuit

Jun 2026 – Present · Virginia, USA

Software Engineer — Financial Services AI · Python · FastAPI · React · TypeScript · AWS

- Own **financial services AI features end-to-end**: discover use cases, execute **technical design**, build Python/FastAPI backend services with **relational databases** and **data modeling**, React/TypeScript front ends for **data-dense financial workflows**, then ship to production on AWS - zero hallucination incidents post-deploy.
- Build **AI-powered features** for financial applications: LLM-backed natural-language query answering, financial summaries, and reporting; ground Claude responses in customer data via RAG so every answer traces to a source record; implement structured-output schemas for consistent, validatable answers.
- Build **production guardrails** around LLM calls: input validation, sensitive-data redaction, output schema enforcement, and fallback paths; LLM evaluation harness with LLM-as-judge, golden datasets, and CI regression detection - caught 3 prompt regressions before release.
- Design **APIs and application state** management for frontend services; containerize with Docker and ship via GitHub Actions CI/CD to AWS with OpenTelemetry monitoring.

Entrata (Apartment & Affordable Housing SaaS)

Oct 2020 – Jul 2024 · 100K+ MAU, 1M+ property records

Senior Software Engineer — Python · Java · React · PostgreSQL · Full Stack · Compliance

- Shipped **20+ full stack features end-to-end** (Python, Java/Spring Boot, PHP, PostgreSQL, React) across 120+ Agile sprints - **concept → technical design → implementation → production** - exercising **product and UX judgment** on **data-dense compliance workflows** for income verification, affordable housing certification, and regulatory filings.
- Owned HUD income-limit ETL and compliance pipeline for 7 consecutive cycles: federal data → 180K **relational database** records with **data modeling**, validation, rollback checks, and compliance enforcement; zero violations - income limit errors affect resident housing eligibility, so correctness is non-negotiable.
- Built automation tools in Python/Bash eliminating 80% of manual operational work; built debugging and data investigation tools cutting triage time 40%; optimized 200+ SQL queries (30% performance gain via execution plan analysis and indexing).
- AWS S3 PHI migration with Libsodium encryption (100% compliance, zero incidents); Docker containerization (50% better execution time); RabbitMQ async processing with idempotency for bulk tenant record updates; reviewed 300+ PRs, mentored junior engineers.

Entrata India (Db Xento Systems)

Sep 2018 – Oct 2020 · Pune, India

Junior Software Engineer — Python · PHP · Java · PostgreSQL · AWS

- Built property management modules and compliance data pipelines processing 1.6M HUD/Tax Credit records with **relational database** schema design and **data modeling**; led AWS S3 cloud migration; created 56 state-specific housing certification documents.

George Mason University — Research Associate / Teaching Assistant

Aug 2024 – May 2026

AI/ML Research · Python · PyTorch · HuggingFace · Statistical Modeling

- Built predictive models (Python, R) achieving $R^2 = 0.977$ across 190+ countries; applied Isolation Forest anomaly detection; built fine-tuned classification model with HuggingFace + PyTorch (presented to faculty panel). Mentored 600+ students.

PORTFOLIO PROJECTS

Leasel — AI-Powered Compliance Investigation Platform

Compliance Automation · Agentic AI · End-to-End Ownership

2026 · Python, LangGraph (8-agent StateGraph), FastAPI, TF-IDF RAG, SQLite, Anthropic/OpenAI APIs · github.com/venkatalakshmiparimala/Lease_complaince_rag

- Owned **end-to-end** delivery of a 3-surface **AI-powered compliance platform** (Document Intelligence & Review, Support Ticket Investigation, Tenant Self-Service) that **automates historically manual compliance workflows** - same value proposition as the target company; **responsible AI enforced in code**: LLM reasons; human or deterministic validator decides every regulated call; safety boundary adversarially tested.
- 8-agent LangGraph StateGraph with real conditional edges and confidence-gated HITL escalation: triage → retrieval → config/history/structured-data agents → reasoning → validation → human; correctly traces false-ineligibility ticket to misconfigured setting tagged to specific release - demonstrates **product/UX judgment** for **data-dense compliance applications**.
- Hand-rolled TF-IDF + cosine reranking RAG with **relational database** backing (SQLite); **LLM-as-judge evaluation**: 93.6% precision, 87.9% F1 across 1,240 test cases; 100% citation correctness, 0% unsupported-claim rate.

AgentProbe — Evaluation Infrastructure for AI Features in Production

Evals · LLM-as-Judge · Full Stack

2025 · Python, FastAPI (15 endpoints), LangGraph (3 agents), ChromaDB, Anthropic, OpenAI, Azure OpenAI, Groq, Ollama · github.com/venkatalakshmiparimala/AgentProbe

- Built 15-endpoint **FastAPI** backend with full **API design**, **data modeling**, and **application state**; routes LLM calls across 5 providers via LLM Router with dynamic task classification; 5-metric LLM-as-judge eval engine (factual accuracy, hallucination, retrieval, completeness, compliance); CI regression detection blocks deploys when quality drops.
- 3 LangGraph agents (Orchestrator, RAG Tracer, Auto Fix) for multi-step evaluation and auto-remediation; used to evaluate Leasel across 1,240 test cases and now integrated into Intuit production AI pipelines.

AI on FHIR — Clinical NLP with Full Stack Architecture & HIPAA Compliance

Full Stack · Compliance

2025 · Python, Flask, React, TypeScript, Docker, BERT, spaCy, LLMs, FHIR REST APIs · github.com/venkatalakshmiparimala/AI_on_FHIR

- **Full stack** NLP application: Python/Flask backend with **API design** translating English/Spanish clinical questions into FHIR API calls using LLMs, BERT, and embeddings; React/**TypeScript frontend** with **frontend architecture** and role-based access control for HIPAA-style data isolation; ~60% reduction in manual lookup time; containerized with Docker.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering — JNTUA, India — 2013–2017

TECHNICAL SKILLS

Full Stack & Web: Full Stack Development, Frontend Architecture, Application State Management, React, TypeScript, JavaScript, Node.js, Redux, PHP, Laravel, C#, .NET, REST APIs, GraphQL, Python (9+ yrs), FastAPI, Flask, Django, Java, Spring Boot

AI & Compliance Automation: AI-Powered Features, Compliance Workflow Automation, Regulatory Compliance Automation, Financial Services AI, LLM APIs (Anthropic Claude, OpenAI, Azure OpenAI, Groq), AI Agents, LangGraph, LangChain, RAG Pipelines, Hybrid Search, LLM-as-Judge Evals, Guardrails, HITL, Prompt Engineering, Structured Outputs

Data & Databases: Relational Databases, Data Modeling, PostgreSQL, MySQL, Oracle, Redis, MongoDB, SQLite, Snowflake, ETL Pipelines, Query Optimization, Stored Procedures, PL/SQL, Data-Dense Applications

Cloud & DevOps: AWS (S3, EC2, RDS, Lambda, Glue, SQS, API Gateway, IAM), Azure (OpenAI, Data Factory, Databricks, AKS), Docker, Kubernetes, Terraform, GitHub Actions CI/CD, RabbitMQ, Kafka, Celery, OpenTelemetry

Retrieval & NLP: Vector Databases (ChromaDB, Pinecone, Weaviate), Elasticsearch, TF-IDF, Embeddings, Semantic Search, BERT, HuggingFace, spaCy, OCR, Document Intelligence

Practices: End-to-End Ownership, Technical Design, Product/UX Judgment, Agile/Scrum (120+ sprints), TDD, Code Review, HIPAA Compliance, PHI Handling, Microservices, Observability